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A Stimulus-Based Model of the Team Adaptation Process: An Integrated Conceptual Review

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As organizations face constant pressures to respond to changing situations and emergent demands, team members are frequently called upon to change their processes and routines and adapt to new ways of working together. In examining adaptation, most researchers have taken a behavior-driven approach where they collapse across the many types of adaptive demands teams face and rely on traditional input–process–outcome frameworks (e.g., Hackman, 1987; McGrath, 1984) to isolate specific behavioral responses. However, this perspective has resulted in several critical limitations. There are key differences in the way teams must collectively respond to different types of adaptive stimuli to be successful, and current research cannot account for or differentiate adaptive demands by stimulus type and needed responses. In this integrated conceptual review, we address these limitations and develop a novel, stimulus-based phase model of team adaptation. We examine studies across our newly developed stimulus detection, urgency identification, and duration assessment phases, and through the team’s adaptive response, adaptive performance, and learning from the experience. We integrate research within each phase of the adaptive process, highlighting factors that demonstrate what successful team adaptation “looks like,” and describe future avenues of research to address key issues within each phase.

Keywords: team adaptation, adaptive stimulus, integrated conceptual review

In modern organizations, teams have become the foundational unit of work and must always be poised to adapt. As teams grapple with constant pressures to adapt to evolving situations and emergent demands, team members must collectively work to adjust their processes and routines to take advantage of new opportunities and overcome emerging challenges. The increasing importance of this adaptability has been driven by the pace of workplace innovation and highlighted by recent exigencies such as the global pandemic, where medical teams have had to adapt to extreme workloads and uncertainty (Hamilton, 2020), while other organizational teams have

been forced to adapt to new work structures (Zeidner, 2020) and unforeseen environmental changes, often with fewer resources and missing team members (Strochlic, 2020).

In an effort to understand how teams can mount effective responses to such changes, researchers have focused on examining the nature and extent of a team’s adaptive response within specific situations (e.g., DeRue et al., 2008; Moon et al., 2004; Randall et al., 2011). Although this approach has yielded valuable insights, the largely *response-driven* focus has led to three critical oversights. First, the focus on team behavioral responses has resulted in a piecemeal approach to understanding team adaptation and the adaptive process. For example, teams must adapt to such diverse situations as environmental or task changes (e.g., Marks et al., 2000), reward structure changes (Beersma et al., 2009), emerging threats (Ellis et al., 2003), team membership and leadership changes (e.g., DeRue et al., 2008; Siegel Christian et al., 2014; Stuart, 2017), communications breakdowns (e.g., LePine, 2003, 2005), and other varied demands. Within each situation, effective adaptive responses have typically only been examined in isolation. This approach has limited the theoretical development of the team adaptation literature, as empirical findings may not generalize well across varied types of adaptive stimuli, and conclusions about effective responses often do not apply to all types of adaptive requirements.

Second, the predominant approach to studying team adaptation neglects the importance of the focal phases of the adaptive process itself. Team adaptation researchers generally have focused each study on only a specific adaptive context, within a single phase, and without recognizing the interdependences and consequences across preceding and subsequent phases of adaptation. This approach has

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resulted in a relatively static view which generally overlooks the critical element of uncertainty, or “incomplete information about the task or the environment” (Argote et al., 1989, p. 59), with which teams must contend, as adaptive demands are often hidden or misunderstood, with their true natures and impact unclear (Baard et al., 2014; Morgeson, 2005; Ployhart et al., 2022). Although some research, such as Bechky and Okhuysen’s (2011) study of special weapons and tactics teams and film crews, has explicitly looked at how teams respond to surprise demands with uncertain natures, magnitudes, and durations, most researchers have controlled uncertainty in their studies to focus on the team’s response. Our model is predicated on the principle that uncertainty within the adaptive environment, and about the specific nature of each adaptive stimulus, must be resolved through a series of collective processes in each phase as the team detects and evaluates each situation to best respond.

Finally, although some researchers have put forth theoretical models of the team adaptive process (e.g., Burke et al., 2006; Maynard et al., 2015), the empirical literature has not typically followed these models in study design, and meta-analytic attempts to examine the overall literature have largely ignored these problems (Christian et al., 2017). Further, because researchers have focused on team responses to distinct adaptive situations and followed traditional input–process–outcome frameworks (e.g., Hackman, 1987; McGrath, 1984), they have also tended to treat team member attributes and team structure as antecedents of adaptation rather than as moderators of the team adaptation process. Although team and situational factors can influence successful adaptation, they do not exert the same benefit for every type of adaptive stimulus, nor do they influence every phase of the adaptive process equally. Rather, we argue that how the team is structured and staffed will play key moderating roles in how well the team can address specific emergent demands at certain points in the process as it unfolds.

Consider an engineering team designing an electronic detection system. Imagine that the team is hard at work designing the device when an overseas conflict causes a looming upstream shortage of a microchip that is integral to their design. First, the team must *detect* the problem—until they do, they will simply continue with their normal routines and fail to recognize that the microchip will be unavailable, losing critical time during which they could adapt. Once they recognize the issue, they must then quickly ascertain how *urgent* the problem is, prioritizing it among other pressing demands, and then assess its *duration*: Can they work around the problem for this particular device, or will this shortage persist more permanently, requiring them to change their entire design? As they implement their adaptive plan and *respond*, they then *learn* through feedback about the effectiveness of their actions, and may then continually update their assessments of urgency and duration as new information arrives. Once the crisis has passed, and old routines are rejoined or new ones refined, the team then aggregates the *knowledge and skills* gained from the experience to apply to future events.

Each of these distinct steps in the adaptive process impacts subsequent phases and will be enhanced or constrained by specific team processes and states, attributes and structure, and the actions of their leaders. The drivers of successful adaptation thus far identified in the literature may only play a critical role in some phases, but have little effect in others, leaving researchers with only general notions of how adaptation unfolds and what factors make teams successful in adapting.

Our integrative conceptual review addresses these issues by proposing a new *stimulus-based* process model of team adaptation to integrate, review, and extend the extant literature. In our model, the adaptation process starts with the team’s detection of an adaptive stimulus in an uncertain environment, followed by the identification and assessment of its relative urgency and expected duration, and then the team’s adaptive responses and outcomes. We also incorporate the critical role of reflexive learning within an adaptive cycle of performance and situational updating, and the experience-based knowledge and skill acquisition that shapes future adaptability, extending previous work that has concluded with the adaptive response.

By expanding our understanding of the adaptation process to include all the critical phases, we advance the team adaptation literature in several important ways. First, we provide a dynamic, integrative theoretical framework that captures the entirety of the team adaptation process, detailing each focal phase. Notably, the stimulus-based perspective allows us to highlight the critical initial phases of detection and urgency assessment, as well as the adaptive learning cycle and learning outcomes, which have generally been neglected in response-based models of team adaptation. Our novel model extends previous frameworks of team adaptation to provide a more targeted understanding of what the team is facing in each phase and how they determine, and learn from, their response.

Second, in offering a comprehensive view of the process, we challenge the limited response-based approach and develop new theorizing for the factors that enhance the team’s ability to adapt within each phase—detection, urgency assessment, duration assessment, response and performance, and learning. We highlight the critical direct and moderating roles of contextual factors within the team, in terms of shared attitudes, orientation, and structure, and external interventions such as leadership and training. Further, we also incorporate nonlinearity between and within specific phases of the model, capturing the complex nature of team assessment and response over time.

Third, we provide specific recommendations and actionable guidance for researchers within each phase, helping researchers to ask important questions and make informed hypotheses within and across phases, then empirically test their models to extend theories and understanding. Within each focal phase, we highlight areas where future research is needed, pointing researchers to critical issues that will allow new theories and findings. We also offer practical steps and targeted actions for managers hoping to improve their team’s ability to effectively identify, assess, and respond to adaptive stimuli and learn from their experiences.

Finally, we discuss empirical challenges and trade-offs in the study of team adaptation, helping researchers better parse previous findings and navigate the hidden pitfalls of conducting team adaptation research in their own work. The varied extant approaches to assessing adaptation have created challenges for researchers in understanding what effective adaptation “looks like” and whether, and how much, positive adaptation has actually occurred. In response, we provide a prioritized agenda for researchers to consider in addressing methodological challenges. By detailing the varied approaches and limitations of each methodological approach, and making specific suggestions for future work, we help researchers to better match their research aims to their design and measurement.

Defining Team Adaptation

A primary challenge of integrating the team adaptation literature is the variety of ways it has been conceptualized by researchers. Team adaptation has been conceived as a capacity of a team, a process in which a team engages, and an outcome of a team's behaviors postchange. For example, Leach et al. (2009) specifically referenced the "adaptive capacity" of a surgical team, such that some teams are more "prepared" to adapt. Most commonly, however, researchers focus on the process of adaptation itself, as shown in the Burke et al. (2006) model and Rosen et al.'s (2011) extension. For example, Kozlowski and Bell described adaptation as a "process of knowledge and skill compilation among team members ... as the team develops its capabilities over time" (2008, p. 16). Finally, some researchers have viewed adaptation through its outcomes, usually in terms of team performance (the *performance change* approach discussed by Baard et al., 2014; e.g., Entin & Serfaty, 1999; Lewis et al., 2007; Marks et al., 2000).

In this review, we include empirical work that has examined team adaptation in all of these forms. We see a clear theoretical connection between (a) *adaptability* as a description of a capacity, (b) *adaptation* as a process, and (c) *adaptive outcomes* as the performance and learning results of adaptation, and we include all these elements in our model, as common to these conceptualizations is a team "dealing with" some type of change. However, we do not argue that adaptation research should be restricted to capacities, processes, or outcomes; the choice to focus on any conception depends on the team's task and the research question, issues which we discuss further in our examination of methodologies for studying team adaptation.

However, we do differentiate team adaptation from two related, yet distinct concepts: (a) continuous improvement and (b) proactivity. Continuous improvement refers to a "company-wide process of focused and continuous incremental innovation" which involves "small step, high frequency, short cycles of change which taken alone have little impact but in cumulative form can make a significant contribution to performance" (Bessant et al., 1994, p. 18). It is commonly studied as the Japanese principle of *kaizen*, which refers to the principle of improvement (Aoki, 2020; Bhuiyan & Baghel, 2005), and is implemented as the never-ending process of continuous improvement (e.g., thinking through how to de-bottleneck processes; Bessant et al., 1994). The key difference from adaptation is that continuous improvement refers to a continued, often company-wide culture, attitude, or mandate to improve processes generally, and not a specific change or stimulus to which a team must adapt.

We also differentiate team adaptation from proactivity, which refers to change-oriented behavior that represents self-initiated and future-oriented action that is meant to actively change the self or situation in an anticipatory fashion—for instance, revising work processes or seeking out opportunities (Hyatt & Ruddy, 1997; Kirkman & Rosen, 1999). Although tangentially related to reactive change or adaptation to emergent demands of the context (e.g., a crisis), proactive changes are distinct in that they modify the situation in a self-initiated way, without a clear requirement to do so (Huang et al., 2014; Shoss et al., 2012).

Scope of the Review

We organize and bound our review based on the theoretical conceptualizations of team adaptation put forth in prior work and

limit our inclusion of studies to those that concern work teams as typically examined in the organizational literature (e.g., "interdependent individuals who share responsibility for specific outcomes for their organization," Sundstrom, 1999, p. 7), excluding work on groups in general (e.g., counseling or affinity groups). Next, we narrowed our search to research that investigated teams faced with some type of adaptive stimulus, only including work that empirically examined how specific organizational or student teams in the context of their work addressed a change or environmental demand.

To identify articles to include in the review, we employed the following: (a) a comprehensive search of the team adaptation literature using the Web of Science database using combinations of the words *team*, *adapt*, *adaptation*, *adaptive*, *resilience*, and *improvisation* in the title, abstract, and topic from years 1900 to 2023; (b) manual searches of relevant major journals (*Journal of Applied Psychology*, *Academy of Management Journal*, *Journal of Management*, *Journal of Organizational Behavior*, *Organizational Behavior and Human Decision Processes*, *Personnel Psychology*, *Organization Science*, *Management Science*, *Administrative Science Quarterly*, *Group & Organization Management*, *Journal of Experimental Psychology*, *Group Dynamics*, and *Small Group Research*); (c) reviewing the reference lists of recent work to locate any articles that did not appear in the initial search; and (d) searching the *Online First* or *In Press* sections of the journals listed above to locate any articles not included in the online database. Following this initial search, articles were eliminated if they did not meet the criteria specified above. Although the initial search led to more than 150 articles, only 91 were retained for inclusion in this review as relevant and illustrative. See Table 1 for a list of exemplar articles included in our review. Transparency and Openness Promotion guidelines related to data sharing, code sharing, hypotheses preregistration, analysis preregistration, and materials sharing are not applicable to this review due to the lack of original data or direct analyses.

A Theoretical Framework of the Team Adaptation Process

We propose a dynamic, learning-based model of the team adaptation process, through which teams cope with the uncertainty of the external environment or changing internal demands by detecting, evaluating, and responding to stimuli. As shown in Figure 1, our stimulus-based model begins with the (1) Detection of the adaptive stimulus itself, and then proceeds through the (2) Urgency Assessment phase, in which stimuli in the environment are prioritized for response. Teams then focus on the most urgent stimulus, using their (3) Duration Assessment to determine their adaptive response. We view those initial phases of detection and assessment as a process embedded in the larger environment through which the team detects, assesses, and prioritizes among multiple demands in the environment. Then, once the team's duration-specific (4) Adaptive Response begins for a given stimulus, (5) Reflexive Learning initiates a recursive cycle in which the team adjusts their response based on learning from feedback, and revisits Phases 2 and 3 as they update their earlier assessments with new information. Finally, in Phase 6, Knowledge and Skill Acquisition, teams compile lessons from their adaptive experience to apply to future events. In the following sections, we

Table 1*Notable Empirical Studies Within Each Phase and Contextual Factor*

Model phase	Exemplar study
Phase 1: stimulus detection	D. Ancona and Bresman (2007); Burke et al. (2006); Christianson (2019); Cross et al. (2000); Denison et al. (1996); A. C. Edmondson (2003); Mayo (2022); Waller (1999); Zellmer-Bruhn (2003)
Phase 2: urgency assessment	<i>Proximity</i> : Christianson (2019); Geiger et al. (2021); Koseoglu et al. (2017); Stephens (2021) <i>Magnitude</i> : Siegel Christian et al. (2014); DeRue et al. (2008); Majchrzak et al. (2007); Ning et al. (2023); Porter et al. (2003); Sommer et al. (2016); Stokes et al., 2010; Zellmer-Bruhn (2003)
Phase 3: duration assessment	D. G. Ancona and Caldwell (1990, 2007); Chen et al. (2005); M. D. Johnson et al. (2006); LePine (2003, 2005); Marks et al. (2000); Porter et al. (2010); Stachowski et al. (2009)
Phase 4: adaptive response and performance	Adelman et al. (2003); Bearman et al. (2010); Beersma et al. (2009); Burtcher et al. (2010); Burtcher et al. (2011); Chen et al. (2005); Siegel Christian et al. (2014); Christianson (2019); A. C. Edmondson et al. (2001); Grote et al. (2010); M. D. Johnson et al. (2006); Kalkman (2023); Kaplan et al. (2013); Lewis et al. (2007); Marks et al. (2000); Marques-Quinteiro et al. (2013); Moon et al. (2004); Munyon and Summers (2024); Porter et al. (2003); Schmutz et al. (2018); Serfaty et al. (1993); Stachowski et al. (2009); Stuart (2017); Summers et al. (2012); Vashdi et al. (2013); Waller (1999); Wiedow and Konradt (2011); Waller et al. (2004); Woolley (2009)
Phase 5: learning	Bresman (2010); A. C. Edmondson et al. (2001); Ellis et al. (2003); Gibson and Vermeulen (2003); Kim and Miner (2007); Konradt et al. (2021); Lewis et al. (2007); Rerup (2009); Santos et al. (2021); Schippers et al. (2014)
Contextual factor	Exemplar study
Team attributes and structure	Abrantes et al. (2020); Beersma et al. (2009); Chen (2005); Chen et al. (2005); DeRue et al. (2008); A. C. Edmondson et al. (2001); Ellis et al. (2003); Han and Williams (2008); Hekman et al. (2022); Hollenbeck et al. (2011); M. D. Johnson et al. (2006); LePine (2003, 2005); Moon et al. (2004); Porter et al. (2003); Porter et al. (2010); Randall et al. (2011); Ployhart et al. (2022); Rapp et al. (2021); Woolley (2009)
Leadership and training	A. C. Edmondson et al. (2001); Entin and Serfaty (1999); Gorman et al. (2010); M. D. Johnson et al. (2013); Koeslag-Kreunen et al. (2018); Marks et al. (2000); Morgeson (2005); Randall et al. (2011); Sanchez-Manzanares et al. (2020); Santos et al. (2021); Serfaty et al. (1993); Sverdrup et al. (2017); Yun et al. (2005)

describe each phase of the model, define key concepts, and employ the model to structure and integrate our review of the team adaptation process and the key contextual factors and boundary conditions that shape it.

Adaptive Stimulus

Although team adaptation is impelled by the emergence of an adaptive stimulus, the initiating nature of the stimulus is rarely highlighted in the literature. This has led leading team adaptation researchers to advocate for a greater emphasis on the role of adaptive stimuli. For example, Baard et al. have noted the need to understand “what it is to which an entity is adapting” (2014, p. 89) and Sanchez-Manzanares et al. have called for a much stronger understanding of the “antecedent side” (2020, p. 3) of team adaptation.

We define an adaptive stimulus as *a demand, problem, disruption, threat, or opportunity that requires the team to purposefully and collectively change its behaviors and/or processes to either mitigate, overcome, or benefit from the situation*. We argue that an adaptive response is “required,” in that ignoring the issue will cause the team to suffer a negative outcome in terms of performance, affect, attitude, viability, expense, or missed opportunities for growth or innovation. For example, failing to change processes when a member is lost will reduce the efficiency and effectiveness of the team, and likely harm the attitudes and affective states of the members. Likewise, missing an opportunity for gain or growth could leave the team at a competitive disadvantage.

This definition relies on work on discontinuous change by Weick and Quinn (1999) and Ning et al. (2023), which refers to a concrete

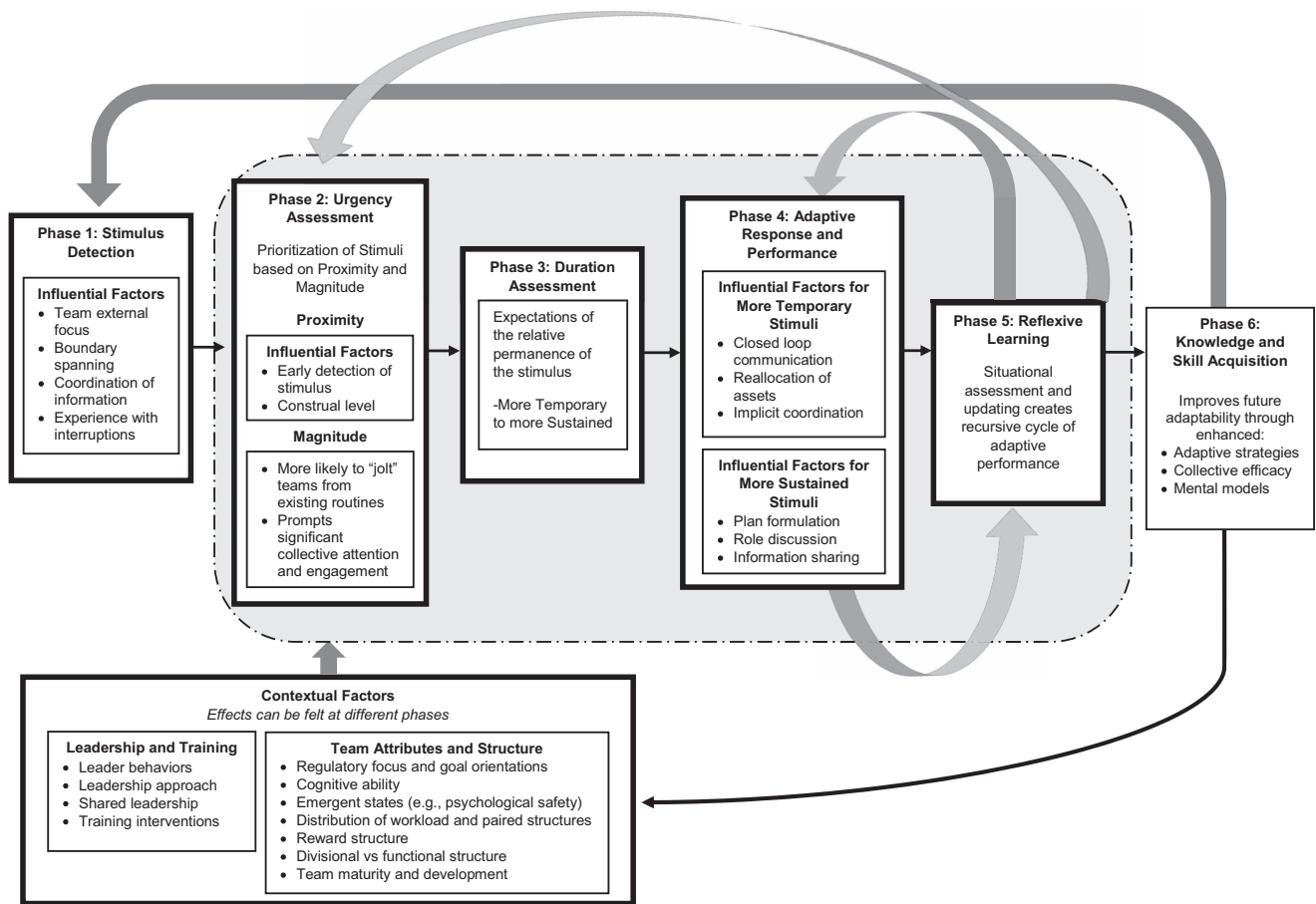
change with a clear starting point or impetus. The set of stimuli within that definition is therefore broad, including such demands as threats and opportunities in the uncertain external environment, experiencing a failure, or changing the structure of the team, but does not include small or routine demands that do not require collective adaptation. As noted earlier, proactive changes initiated by the team without a specific impetus or demand do not fit this definition. However, although the extant literature is primarily focused on threats and adverse demands, as we will discuss in our discussion of *Knowledge and Skill Acquisition* in Phase 6 or our model, even negative events may ultimately lead to positive change if the team learns from the experience. Our focus on stimuli at the front end of our model allows for a fuller examination of the process by which teams adapt and a much-needed exploration of how the team reacts, plans, and makes decisions in response to the specific stimulus throughout the adaptation process.

Stimulus Detection

Phase 1: Stimulus Detection

The team adaptation process begins with the initial detection of an adaptive stimulus. Uncertainty is a defining feature of organizational contexts (Katz & Kahn, 1978), and teams must be able to recognize within their environment which demands require a response (adaptive stimuli) and which do not. How quickly that detection occurs will ultimately determine both the nature and effectiveness of the team’s adaptive response in later phases of the adaptation process (Gersick & Hackman, 1990). Previous theory has emphasized the

Figure 1
A Stimulus-Based Phase Model of the Team Adaptation Process



initial recognition of cues, or streams of cues, related to a stimulus as the precipitating factor that begins the team adaptive process (e.g., Burke et al., 2006). However, very little work has directly examined the detection process or identified factors that enable or impair teams' detection of adaptive stimuli.

Once a potential environmental demand, disruption, opportunity, or threat is noticed by one or more members of the team, they must individually or collectively determine whether it reaches the level and definition of an adaptive stimulus as requiring the team to purposefully change its behaviors and/or processes to match each unique team context. Based on member experience and the nature of the team task, a member may either voice the concern to the team or refrain from raising the issue for routine problems or novel, but small, demands that will not significantly impact the team. For example, a member needing to take a bathroom break in a software development team likely will not warrant notice or response, but such a break during surgery may require a temporary shift in a surgical team's roles or actions. Although the stimulus detection process may often depend on the voicing of a concern or issue noticed by a team member to the rest of the team, such actions are reactive to a specific stimulus rather than being self-initiated, proactive changes (Huang et al., 2014).

Uncertainty in the environment, and the varied nature of adaptive stimuli, presents challenges for team members' limited attentional

resources (Rerup, 2009; Weick & Sutcliffe, 2006), resulting in delays in the detection and identification of many stimuli as actual demands. For example, teams may not notice a subtle change in their environment, such as a plane's autopilot becoming disengaged and the aircraft gradually losing altitude (Helmreich & Foushee, 1993). Even planned organizational interventions assume that a team will always recognize and respond to the new demand, even though teams may miss a poorly communicated change or requirement. For example, a university IT team may not notice a change in school policy, leading them to invest time and resources in a course of action that is outdated. Sometimes, uncertainty is worsened by too much competing information, drowning out important cues. For example, in her 1999 study, Waller studied flight crews who faced a simulated emergency hydraulic failure. She added contextual factors of bad weather and a noisy descent which made the cause of the actual problem uncertain and difficult to detect, resulting in many crews missing or failing to verbalize the true threat.

How rapidly a stimulus is detected will help determine how quickly the team will be able to assess and prioritize it in the subsequent Urgency Assessment phase (see the Stimulus Detection: Future Directions section) and be able to respond to the most urgent stimulus in a timely fashion. If detection is delayed, the team's options become limited, and the team may need to preempt other urgent stimuli and suboptimally scramble to respond to the new

priority. For example, Christianson (2019) found that more effective medical teams regularly monitored for problems and attended to salient cues more quickly (e.g., noticing a patient's chest was not moving with bagging). In contrast, less effective teams suffered delayed cue detection or missed cues entirely following an emergency.

Early stimulus detection may be particularly important for initially minor or hidden issues that slowly accumulate into eventual problems or even disasters. For example, investigations of high-profile organizational failures (e.g., Union Carbide Bhopal, BP Deepwater Horizon) have found that those disasters stemmed from teams neglecting small issues that gradually accumulated to degrade safety and precipitate larger problems, leaving teams without enough time to respond (Shrivastava et al., 1988). Numerous systematic failures have similarly been attributed to frontline teams discounting the substantial long-term dangers of their endeavors while focusing on the short-term operational requirements (e.g., Weick & Sutcliffe, 2015).

Researchers have identified several factors that may help a team be more attuned to noticing stimuli early enough to optimally respond. Some work has suggested that an *external focus* might be beneficial. For example, externally oriented teams (e.g., D. Ancona & Bresman, 2007) have been shown to enable earlier detection of threats and opportunities in the team's environment. Similarly, *boundary-spanning activities* have been found to enable teams to better sense adaptive demands in the environment (e.g., Cross et al., 2000; Denison et al., 1996; A. C. Edmondson, 2003). Recent research has also identified the importance of *coordination of information* throughout the team and the *inclusion of peripheral members*. Mayo (2022) found that teams of medical professionals were more likely to notice cues when they coordinated or interacted with peripheral members of the team, leading to less wasted time and better care. Mayo (2022) also found that *live discussion* within the team was more effective than virtual communication in helping the team detect relevant cues. Similarly, Waller (1999) highlighted the importance of *nonroutine event verbalization*, an indicator of stimulus detection, and also found that crews that immediately engaged in task prioritization and distribution were more effective. Counterintuitively, Zellmer-Bruhn (2003) found that *interruptions*, which may harm efficient performance, may also create a "cognitive pause" that makes the team more receptive to picking up changes to the environment, even without an active search.

Stimulus Detection: Future Directions

Ideally, all stimuli are detected early enough to be fully considered in the team's Urgency Assessment phase. However, if teams focus too much on detecting cues, they risk "lowering the warning threshold" (Parker & Stern, 2002, p. 622), and spending time and resources attending to false alarms along with actual demands. Adaptation research has thus far generally focused on situations in which teams are required to respond to changes to be successful. However, future research should examine teams' ability to correctly identify whether a detected stimulus requires an adaptive response. Work is needed to identify team attributes that may be helpful for stimulus detection, such as team member individual experience and team tenure; contextual factors (e.g., training), or building on Zellmer-Bruhn (2003), we may find that interventions such as "reflective pauses" or flowcharts of common scenarios can help teams determine whether a stimulus requires a coordinated team response.

More research and theory building are also needed in the early stages of team adaptation, as teams that are best able to detect stimuli may need a combination of both an internal and external focus (e.g., D. Ancona & Bresman, 2007), and an ability to coordinate with peripheral members and external contacts to detect stimuli more quickly. One avenue may be through structuring teams to include boundary-spanning roles (e.g., D. G. Ancona & Caldwell, 1992), but there may be other factors that help teams to develop both an internal-external focus and to include periphery members. For example, future work may find a role for status differences in teams and a culture of limited hierarchy in an organization, motivational factors, leadership behaviors such as *taking charge*, or shared planning behaviors such as *anticipating* or imagining future states together (Grant & Ashford, 2008; Morrison & Phelps, 1999).

Teams may also face potential occasions for gain that pass by without recognition, and such missed opportunities remain an unstudied area. As we discuss in our contextual factors section, some research suggests that a team promotion focus (P. D. Johnson et al., 2015) might attune the team toward looking for opportunities. Yet currently, we know little about how and why teams take advantage of, or miss out on, stimuli that present opportunities for gain.

The Adaptive Evaluation and Performance Cycle

Once an adaptive stimulus has been identified, it must be assessed, and the team must formulate an effective response based on its expected duration assessment to behaviorally adapt. As the team's adaptive response unfolds, and they receive feedback and reflect on their effectiveness, they may update their initial assessments of urgency and duration, and reshape their adaptive response over time. As shown in Figure 1, we view that process as a recursive performance and learning cycle, constituted within Phases 2–5, followed by the aggregation of knowledge and skills acquisition from the resolution of adaptive episodes in Phase 6.

Phase 2: Urgency Assessment

Once a stimulus has been detected as a potential problem, threat, or opportunity, the team must next determine its relative level of urgency within the context of all of the routine and novel demands a team faces at a point in time. We define *urgency* in terms of the team's perceived need to prioritize their response to a stimulus based on a combined assessment of both its perceived proximity and magnitude. Both factors play a critical role in how the team assesses a stimulus; *magnitude* is the perceived size or scale of the stimulus (how large will the consequences be?), and *proximity* is the relative temporal closeness of the demand (how soon before those consequences will be felt?). The team must jointly assess and balance those dimensions, determining an overall level of perceived urgency to inform their prioritization of each stimulus response. For example, a threat of moderate magnitude, but high proximity, may be deemed more critical than a larger demand further in the future.

Although researchers have called for more investigations into how teams and leaders can deal with crises (James et al., 2011; Sommer et al., 2016), studies directly comparing relative levels of urgency are limited. Indeed, most laboratory studies tend to examine only a single adaptive stimulus with the same characteristics for all teams, and field studies tend to be retrospective, making

it difficult to discern how urgency assessment unfolded in the moment. However, findings thus far suggest that greater urgency acts as a powerful spur for a quick reaction to a stimulus. For example, in a study of firefighters, Geiger et al. (2021) found that firefighting teams looked to urgency cues to determine how quickly to respond, from abandoning a meal in progress for an urgent situation (a car accident with potential casualties) to chatting while changing out of sweaty clothes before mobilizing for a less urgent alarm (cat on a rooftop). Determining the urgency of a stimulus also shaped firefighter behavior on the way to the incident; for a nonurgent situation, the team drove more slowly to avoid endangering the public and themselves.

Some stimuli may require an immediate response while others are distal enough to provide time for discussion and planning or the prioritization of other tasks. The ability of the team to properly evaluate and balance those dimensions will determine how effectively they prioritize and respond to them. Information processing theory suggests that teams with high task interdependence, such as those with specialized roles or expertise, will tend to engage in higher quality information sharing, enabling them to develop better cognitive maps of the demands currently confronting the team (Hinsz et al., 1997) and therefore prioritize more accurately.

Some critical team roles may overly influence which stimuli should be prioritized due to their position within the team's workflow (Crawford & LePine, 2013; Humphrey et al., 2009). This may be especially true for teams with prevention-focused leaders, as the leader's regulatory focus may play a key role in prioritizing stimuli for their team through setting goals and monitoring their actions, as the magnitude of losses may weigh much heavier than for leaders with a promotion focus (Kark & Van Dijk, 2019).

However, the process is not purely cognitive, as the emotional response of the group to the stimulus may play a key role in its urgency-based prioritization. Stimuli viewed as urgent create a visceral reaction and emotional response (Stephens, 2021), leading to a greater imperative to quickly respond. Emotions act as a signal or "alarm bell" that can signal to the team a higher need for change. We discuss how teams' assessment of each dimension of urgency—proximity and magnitude—determines their overall prioritization of each stimulus below.

Stimulus Proximity

Proximal demands or threats tend to be viewed as especially urgent relative to distal events, as the construal level of proximal events makes them seem much more concrete and actionable (Trope & Liberman, 2010). Conversely, distal events will tend to appear more abstract and less "real" or vivid, making them feel less urgent and therefore harder to envision and act upon (Vogus et al., 2010). Distant events are also inherently more uncertain, with their expected arrival and timelines less clear. Unfortunately, due to this tendency, when a stimulus eventually does become more proximal—after initially being viewed as "distant"—and finally becomes prioritized by the team for a response, they may have missed their best opportunity to effectively plan for how to engage with the stimulus.

Other work has shown how perceptions of proximity can influence downstream team processes and outcomes. Koseoglu et al. (2017) found that teams experiencing a deadline change later in their work cycle exhibited lower levels of creativity than teams that did not experience the change, or experienced it earlier.

Christianson (2019) found that when a situation is highly urgent—a patient needing immediate medical care—the most effective teams were able to simultaneously search and test explanations and assess the situation while noticing attributes of the situation and gathering information. Such research highlights the potentially negative effects of unexpected changes to proximity and underscores the need to develop more research on connections between the stimulus detection and urgency assessment phases.

Stimulus Magnitude

Significantly impactful or threatening stimuli will be more likely to garner significant collective attention from the team. However, adaptive stimuli with seemingly less impactful potential consequences may result in less emphasis on effective adaptive responses from team members and leaders, and a lower prioritization of the response. Foundational theory addressing habitual routines in groups emphasizes this difference between minor and major stimuli, and suggests that extreme or truly novel situations will tend to be more likely to "jolt" a group from its existing patterns (Gersick & Hackman, 1990). Supporting this idea, Zellmer-Bruhn (2003) found that greater interruptions led to greater collective efforts toward knowledge acquisition and transfer, and Porter et al. (2003) found that high-magnitude demands (e.g., an extreme influx of threats) led to more help when the help needed was perceived to be legitimate. Similarly, both DeRue et al. (2008) and Siegel Christian et al. (2014) examined how teams adapted to the loss of members, finding that the loss of more "critical" members of the team prompted the remaining members to engage more quickly in behavioral change. Thus, higher magnitude stimuli may be more effective at prompting teams to adapt quickly, by prompting teams to "regroup" and focus on their *task capability redevelopment* (Ning et al., 2023). Such rapid responses may be particularly valuable in helping teams take advantage of new opportunities before their competition.

Urgency Assessment: Future Directions

An important question revolves around how teams determine urgency in the context of multiple simultaneous demands. Thus far, we know of one article to date (Stokes et al., 2010) that has examined the ability of teams to prioritize among multiple stimuli (in that case, a task change followed quickly by a communications breakdown) and handle any potential conflicts. For example, teams may inadvertently make adaptations in response to an emergent stimulus that leave the team less able to respond to a second stimulus occurring soon after. Future studies should examine how teams might react to changes that co-occur, such as in a team that is downsized at the same time it finds out it will be losing funding for a project. Although an overall effective response might be an additive combination of the different behaviors, multidimensional adaptive stimuli might require either prioritizing between stimuli for adaptive response or developing a unique set of behaviors and skills. Future research should examine such complex, yet realistic, situations.

Another key limitation of the extant literature in terms of urgency assessment is that most researchers have only studied one stimulus at a time, such that we cannot directly compare prioritization of the same type of stimulus with a lower or higher level of severity. Work that specifically addresses how teams can engage in plan formulation, shift their roles, or identify attributes and structures

most important for higher versus lower urgency stimuli would be valuable. Further, team affect may play an understudied role in assessing stimulus urgency. For example, Kaplan et al. found that too much positive affectivity in a team “may result in the team’s collective failure to appreciate the potential severity of the crisis” (2013, p. 486), which would likely hinder productive response patterns.

Christianson’s (2019) results provide more support for the idea of team *sensemaking* around urgency and how to assess urgency for extremely pressing situations. In situations with extreme urgency, teams likely will not have time to plan and discuss before acting, in which case the most effective teams appear to sense-make while they act and engage in continuous information updating. In situations of more moderate urgency, teams may still have time to come up with a coordinated plan of action, but we know little about which team-level factors or inputs will result in more accurate urgency assessment or planning during this phase. Our model suggests that an *external focus* (as we identified in Phase 1) may be an area to consider, as it helps teams detect stimuli, in effect reducing proximity during Phase 2.

Although stimuli can be opportunities as well as problems or threats, the extant literature has overwhelmingly tended to focus on negative events rather than potentials for innovation, gain, or competitive advantage. Although potential losses may loom larger than gains for teams and their leaders, missed opportunities may end up costing the team more in the long run than solving small problems. Research focused on helping teams better recognize the potential magnitude of the effect of opportunities, and more accurately prioritizing their pursuit, is much needed.

Finally, most studies have tended to focus on stimuli with a relatively low or moderate magnitude, with little work focusing on teams adapting to “extreme” contexts with severe stimuli, such as Sommer et al.’s (2016) investigation of a widespread hospital bed shortage and Majchrzak et al.’s (2007) study of emergency response groups during Hurricane Katrina. In such cases, researchers found that different factors may emerge as more critical than in other settings. For example, Majchrzak et al. (2007) suggested that experience-based credibility becomes less important for coordination than members’ ability to create “swift trust” (Schilke & Huang, 2018) for high-magnitude stimuli. More work is needed that examines settings of extreme urgency.

Phase 3: Duration Assessment

Although the first two phases of the model examine all the demands within an environment to determine which stimuli should be prioritized, the subsequent phases focus on the assessment and response for each specific stimulus and its response. The team’s best response to any stimulus will largely be a function of their assessment of its duration. How long the stimulus is expected to cause an effect helps the team pinpoint the best type of response to fit the situation—a short-term workaround versus a longer term adaptation. As Kozlowski et al. noted, teams must “reconfigure their workflows and modify their performance to satisfy short- and long-term demands for change” (1999, p. 253). For example, a team dealing with a critical member going on vacation for a week will, and should, adapt very differently than when dealing with the permanent loss of a member without replacement. Similarly, a single day-long loss of communication between virtual team members

requires a very different response than the loss of a reliable network connection for the foreseeable future.

Critically, more temporary stimuli are best understood not by the actual duration (i.e., minutes, hours, days, weeks, or years until the stimulus disappears, if it ever does), but rather by the expectation of team members that the situation will return to the way it was before the change. The true duration may be uncertain (e.g., a team member out sick with flu), and teams must respond based on their current expectations and respond accordingly, “making do by applying combinations of the resources at hand to new problems and opportunities” (Baker & Nelson, 2005, p. 331). Teams facing more temporary stimuli, such as power plant crews facing novel events (Stachowski et al., 2009) or teams with a short-term workload imbalance (Porter et al., 2010), may be most effective if they opt to develop rapid, makeshift “workarounds,” for example, short-term communications adjustments (LePine, 2003, 2005).

In contrast, more sustained or relatively permanent stimuli require the team to respond to the updated situation as the new normal, such as a change to the team reward structure (M. D. Johnson et al., 2006), or a more permanent change in the team’s operational environment (Chen et al., 2005; Marks et al., 2000). In these conditions, teams tended to be more successful if they invested resources to thoughtfully develop plans to transform their task work and cognitive structures to incorporate new routines. Previous work in this area has taken a categorical approach to this issue, such that stimuli have been classified as either temporary or permanent (e.g., Christian et al., 2017). However, relying on simple classifications may cause researchers to miss critical differences across relative levels of permanence and leave important gaps in our understanding. Some stimuli may be best viewed as “somewhat temporary” or “more sustained, but not permanent” leading to very different attitudes and responses than to other nominally similar durations. Therefore, we take a more continuous view of duration in our model.

An accurate assessment of a given stimulus’s duration is a key factor in how well an adaptive response can be designed and executed, which is why our model places duration assessment not as a contextual moderator, but rather as a key phase that precedes, and informs, the team’s adaptive response. Until the team has assessed how long they expect the stimulus to be present or continue to affect the team, they cannot develop an optimal response. The issue of time and the assessment of duration are particularly important for more sustained stimuli where an effective adaptive response in the long term may have short-term performance costs. In these cases, the relationship between collective behavior and performance may not be obvious and may not manifest immediately, and many effective adaptive responses in the long term may have short-term performance costs. Research has shown strong connections between duration assessment and the effectiveness of behavioral responses which we elaborate on in our *Adaptive Response and Performance* phase.

Duration Assessment: Future Directions

Accurate assessment of stimulus duration is critical, as incorrect expectations can result in misaligned responses, inefficiencies, and poor adaptive performance. Unfortunately, researchers have yet to address the important question of how teams can best assess stimulus duration, pointing to a critical direction for future research. In many cases, teams should be able to reliably assess when and whether a stimulus will end or when things will revert to the

previous situation, especially if the team has experience and knowledge of their organizational landscape. We suggest that other factors influencing the accuracy of the duration assessment are high levels of communication within the team and with managers, other organizational leaders, and other stakeholders. As an even more specific form of communication, we also argue that *feedback-seeking* (e.g., Sherf & Morrison, 2020; Zhang et al., 2022) should be helpful, as teams that seek out feedback about their performance and plans are likely to learn more about the “fit” of their approach. Although researchers have shown that general communication behaviors and certain specific forms of communication are helpful for the adaptive process in general (e.g., Christian et al., 2017), future work should investigate the role of feedback-seeking and potentially build it into future phased-based models as a critical component in duration assessment.

We further suggest that boundary-spanning behavior—such as being an *ambassador* or *scout* (D. G. Ancona & Caldwell, 1990)—should allow for more accurate duration assessments, as teams that engage in higher levels of these behaviors should be more informed about the external environment and shocks to their system like new regulations, and whether these factors will be permanent in duration. Team structure may also be a key factor in accurate duration assessments, as interdependence naturally forces interaction (Hinsz et al., 1997), which may result in more accurate inferences. If team structures or roles cannot be adjusted, we also theorize that the intentional structuring of reflective action sessions into teamwork activities (e.g., Vashdi et al., 2013) could promote pattern recognition and better assessment of the duration of stimuli.

However, we do note that highly uncertain environments and unique situations—such as the COVID-19 pandemic—present such an unknown challenge that even the most experienced teams may operate under misperceptions about the duration of the stimulus. If a potential problem is viewed as temporary, but persists beyond that initial expectation, the team’s response may be ineffective, forcing the team to reassess the situation and potentially have to adapt again. If the problem is expected to be permanent, but actually resolves more quickly, significant effort may be wasted in changing processes, then having to revert back again. Therefore, although this phase generally occurs in a straightforward, linear fashion, we argue that effective teams will engage in updating in the *Reflexive Learning* phase (Phase 5), which we will describe in greater detail. In the future, researchers should consider exploring higher uncertainty situations and examine this phase as a cycle of error identification and correction based on a misdiagnosis of duration and how the team must revisit this phase once more information emerges.

Phase 4: Adaptive Response and Performance

The team’s adaptive response encompasses team members’ collective behaviors toward one another and their shared task as they attempt to respond based on the expected duration of the adaptive stimulus. Team adaptive responses enumerated in previous models and meta-analyses fall into two major categories: processes and emergent states (e.g., Burke et al., 2006; Christian et al., 2017). Researchers have varied in their examination of this topic, some focusing on the mediating role of collective team behaviors and states, but others examining adaptive outcomes such as performance directly. Theoretically, behavior and outcome are inextricably

linked because the latter follows the former. Therefore, whether behavior or outcome better captures the nature of adaptation may be best determined by the researchers’ time horizon. Consider a research setting involving teams adapting to a sustained change with long-term outcomes. In such cases, the team’s immediate process changes may not result in significant, measurable adaptive performance improvement until weeks or months later. For that situation, adaptive behaviors may be a more useful indicator of effective adaptation in the short term.

Below, we review the literature, organizing research findings by relative stimulus duration, to illustrate the relationship between stimulus duration and their most effective responses. Notably, some behavioral responses (e.g., general communication, support behaviors) have been shown to be effective for both more temporary and more sustained stimuli.¹

Adaptive Responses Most Effective for More Temporary Stimuli

The most studied team process for adaptation is *communication*. For example, metaqueries and responses or closed-loop communication (communications that are transmitted, acknowledged, and then verified as correct by the sender) have been shown to relate to greater shared awareness and performance in teams facing temporary adaptive stimuli (Bearman et al., 2010; Walker et al., 2009). Various forms of more short-term or “in the moment” team coordination and communication have also been shown to be important: for example, *task and information management behaviors* (Adelman et al., 2003; Burtcher et al., 2010, 2011), *information collection and transfer* (Waller, 1999), and *backing up behavior*, especially when a member has a legitimate need for help (Porter et al., 2003). *Relational experimentation*—trying out new combinations of members for task accomplishment (Stuart, 2017)—and *reallocation of assets* are also key, as teams are able to respond to an incongruent mission scenario by reassigning assets to address an opponent’s new strategy (Serfaty et al., 1993). Similarly, *bricolage*—making do by applying novel combinations of resources when facing a problem—for example, by shifting roles or routines—helped organizational teams adapt and better draw on shared resources when facing surprises (Bechky & Okhuysen, 2011).

In addition, there has been some support for the role of behaviors related to shared cognition in adapting to more temporary stimuli. For example, *mental model updating* to include new cues (Christianson, 2019) has received support in predicting adaptive performance, and Marques-Quinteiro et al. (2013) found that engaging in team *transactive memory coordination* behaviors leads to more adaptive outcomes.

However, to fully realize the benefits of coordination in temporary adaptive situations, some research suggests that teams must also shift from explicit to *implicit coordination* (Rico et al., 2008), although there are mixed results. Stachowski et al. (2009) found that higher performing nuclear power plant control room crews employed simple interaction patterns, engaging in shorter and

¹ It is also important to note that, to our knowledge, none of these studies have explicitly assessed the team’s assessment of duration. Instead, researchers have created conditions in which teams are either explicitly informed of duration or assumed to make an accurate assessment. We note this issue as important for future research in our discussion.

more concise interactions. This indicates both a reliance on implicit coordination processes and an ability to shed established patterns of interaction when faced with a stimulus. However, other research has found a greater benefit in temporary situations for explicit, heedful interrelating types of communications (where members constantly reconsider the consequences of their actions on the goals and actions of others; Grote et al., 2010) and *explicit face-to-face communications* (Waller et al., 2004). Marques-Quinteiro et al. (2013) explained these contrasting results by showing that the effectiveness of implicit coordination hinged on the team utilizing a transactive memory system—teams that lacked strong systems adapted less effectively through implicit coordination and needed more explicit direction.

Although not a behavioral response, team affect may play a role in adaptation to temporary stimuli. In a study of teams engaged in a simulation where they experienced a temporary stimulus (nuclear power plant crews responding to temporary nonroutine events), Kaplan et al. (2013) found that the team's experience of negative emotions directly hindered adaptive performance in response to the temporary crisis. A recent theory article also focused on the role of emotions in directing and motivating adaptive behaviors, especially as "pathos appeals" from leaders, such that the team's emotional state can enable new adaptation options in the face of a temporary challenge (e.g., unfair referees; Munyon & Summers, 2024).

Future Directions for More Temporary Stimuli

The research above suggests a benefit from quick communications, role reorganization, shared cognition, and engaged emotions when facing more temporary stimuli. Despite these findings, there is clearly room for more research in this area. For example, for teams facing short-term adaptive stimuli that must develop efficient "workaround" solutions, creative thinking may play an important role in adaptation. Future research focused on composition models may also provide an understanding of how the distribution of creativity within a team relates to adaptation (see Chan, 1998; Kozlowski & Klein, 2000). Perhaps it is only necessary that one member of a team think creatively and suggest a solution to an emergent issue (a disjunctive composition model), resulting in adaptive behavior.

Adaptive Responses Most Effective for More Sustained Stimuli

Although communication is also key for more sustained stimuli, research suggests that different types of communication may be more important when adapting to more sustained changes. In contrast to the quick "question and response" or closed-loop communications found to be effective for more temporary stimuli, more involved types of communications appear particularly effective for stimuli leading to longer term demands and changes. Time to discuss and reflect on the change is helpful for more sustained changes, as these types of longer term changes require teams to come up with a plan for reprioritizing tasks and reconfiguring their strategy. For example, *plan formulation*—deciding on a course of action, setting goals, and clarifying member actions (Burke et al., 2006)—is a key factor for positively impacting adaptive performance (Siegel Christian et al., 2014).

Similarly, *action identification behaviors* (active discussion and identification of actions; Woolley, 2009) positively impact team adaptation to sustained stimuli. *Role discussion* helps to buffer the challenges of more permanent reward structure changes and enables teams to coordinate better, thereby boosting performance (Beersma et al., 2009), and *team reflection* has been found to be particularly helpful across contexts (A. C. Edmondson et al., 2001; Lewis et al., 2007; Schmutz et al., 2018; Vashdi et al., 2013), although one study found that reflection was negatively related to coordination and perceived team performance (Wiedow & Konradt, 2011).

Along the lines of more in-depth communication processes, *between-episode regulatory processes*, such as goal striving and team transition processes, relate to adaptive performance for sustained stimuli (Chen et al., 2005). More general *communication behaviors and processes* and *information sharing* also positively relate to adaptive performance for both sustained and more temporary stimuli (M. D. Johnson et al., 2006; Summers et al., 2012), and also impact team mental models and adaptive performance for more sustained stimuli (Marks et al., 2000). Finally, research also finds that *support* (providing direct help to a team member) is an effective behavioral response for longer term stimuli as well (Moon et al., 2004), which parallels the results for backing up behavior in response to more temporary stimuli.

Future Directions for More Sustained Stimuli

Given the increased interdependence of work teams in organizations (cf. Cohen & Bailey, 1997), it is important to understand how teams successfully adapt to greater coordination and centralization, which highlights an important area for future research. For example, individuals may work in cross-functional teams, where they likely have conflicting goals or misaligned incentives (e.g., Pearsall & Venkataramani, 2015). Research is needed that investigates which types of interventions and behaviors will allow individuals to successfully adapt to a highly interdependent team environment. Further, in terms of team composition, sustained stimuli often involve more permanent alterations to team membership; thus, newcomer attributes and the relationship that the newcomer has with current team members may be another important predictor of adaptive performance.

Another question for future research involves additional factors that allow teams to break from old habits. Our review points to various types of team training and both directive and supportive leader behaviors as particularly helpful (see the *Contextual Factors: Leadership and Training* section). Team training appears to most directly impact communication and shared cognition, and leaders may play a crucial role in responding to more sustained stimuli as they can help a team to understand their new reality, abandon old routines, and focus on the future. For example, mission briefings may be most effective for more temporary stimuli, whereas in-depth training and supportive coaching may be necessary for more sustained stimuli.

Team Learning

Learning changes the way the team understands and interacts with its environment (A. C. Edmondson et al., 2001). Successful, and often unsuccessful, adaptive experiences can provide an opportunity for team learning and growth (Rerup, 2009), and researchers have

suggested that learning is an important outcome of the adaptive cycle (Kozlowski & Bell, 2008). Learning can therefore enable both short-term and long-term changes, providing immediate information to enable adaptive changes for a specific stimulus during performance, as well as growth and improvement as teams learn from each adaptive episode, enabling them to apply that new knowledge and skills to detect and respond to future adaptive stimuli. In our model, as shown in Figure 1, we differentiate between the former as a crucial part of the adaptive performance cycle with a specific stimulus response (*Reflexive Learning*) and the latter in terms of broader growth and development following an adaptive episode (*Knowledge and Skill Acquisition*).

Phase 5: Reflexive Learning

Within a specific performance episode, team learning is “an ongoing process of reflection and action, characterized by asking questions, seeking feedback, experimenting, reflecting on results, and discussing errors or unexpected outcomes of actions” (A. Edmondson, 1999, p. 354). Teams learn from trying new things, having successes, or making mistakes, and then jointly reflecting on choices and actions (Ellis et al., 2003). This learning process allows them to make changes “on the fly” to tailor their adaptive response (Kozlowski & Bell, 2008), creating a recursive cycle of adaptive response and learning from the effectiveness of that response. Researchers refer to this “within episode” process as *reflexive learning* (Harvey et al., 2023) and have shown that cycles of learning and performance during a task or project allow for more effective adaptation of strategies and procedures (e.g., Konradt et al., 2021; Schippers et al., 2014). For example, Ellis et al. (2003) demonstrated how teams faced with new “unknown” threats had to experiment to learn how to respond to them, and then collectively share that information within the team, to enable members to adapt their tactics during the performance episode.

We position reflexive learning following the team’s efforts to adapt to each stimulus based on their urgency-based prioritization and duration assessment. As team members collectively make changes to their processes and interactions, reflexive learning begins with their evaluation as to whether they have successfully begun to adapt. The team looks for feedback about the effectiveness of their actions, as implementing decisions causes the team to reassess temporal issues as the implications of their choices emerge (Kaplan & Orlowski, 2013). Such feedback may be internal, as they evaluate their success based on their feelings of reduced stress or enhanced capacity, or external, based on their achievement of some goal or effectiveness criteria. Initially, although the team may have engaged in an ultimately effective response, they may still suffer an early performance decline from changing routines, which takes time to resolve. Eventually, the team (and/or their leadership) will either have to appraise the adaptation as successful and curtail further changes, or new demands will emerge that supersede the stimulus and force the teams’ attention elsewhere.

We propose that this constant search for, and response to, feedback kicks off two reflective processes: updating earlier phases of the team adaptation process to reevaluate assessments of urgency and duration and making microadjustments to current adaptive behaviors. Due to high levels of uncertainty in the environment and about new stimuli, the teams’ initial evaluations in earlier phases

may be inaccurate, and as they begin to update their information, they may need to make adjustments. As Geiger et al. (2021) noted, the temporal fit between internal routines and the external environment does not happen simultaneously—it requires multiple iterations, gradually fitting the rhythm of the routine to the nature of the external demand. This reflexive learning process results in effective *updating*, which refers to “revising provisional sensemaking to incorporate new cues” (Christianson, 2019, p. 45), helping resolve uncertainty through multiple cycles of performance and reflection.

During the initial Phase 2 assessment, a stimulus may initially be viewed as less urgent, either too temporally distant or too low in magnitude to prioritize, leading to a slow adaptive response, but over time the true picture should become clearer. For example, some threats, such as an approaching storm expected to miss a boat at sea, may change its track or suddenly become more powerful, forcing the crew to reassess both its proximity and magnitude and subsequently reprioritize their response. Doing so may cause them to shift their focus away from other adaptive demands, causing a cascade of changes within, and by, the team. Relatedly, the team’s initial assessment of magnitude may persist even in situations where the potential magnitude of a demand is growing, causing the team to leave a rising threat underprioritized and limiting the effectiveness of their eventual response.

Similarly, in Phase 3, the duration of a stimulus may change or become clearer through reflection, highlighting any incongruities between the updated situation and the team’s no longer effective response. For example, a team member may initially be gone on what is expected to be a short sick leave or travel and then is unable to, or opts not to, return at all. The team’s effective short-term workaround for dealing with being shorthanded would no longer make sense, leading the team to reassess and revise their adaptive strategy as they work through the Duration Assessment phase for a second time. In addition to the possibility of a change in duration more directly, as discussed above, members may also make mistakes in duration assessment. In terms of “course correcting” following a misclassification of duration, we recommend that future work examine team states such as psychological safety in the role of helping teams readjust following this type of mistake, as psychological safety has been shown to help teams learn from setbacks and mistakes (e.g., A. Edmondson, 1999).

In Phase 4, the adaptive response and performance, even well-matched adaptive strategies may be less successful than expected and the team will need to make adjustments to its processes. For example, a team reallocating tasks among remaining members as they cope with a missing teammate may realize that their efforts are suboptimal and may recognize the need to reassign tasks based on varied member skills or emergent changes in some members’ workload that cause a bottleneck in workflow. Other times, if actions are not going as planned and not having the desired effect, or something else has changed to make the previous planning obsolete, teams may have to engage in *bricolage*, applying novel combinations of resources to address a new problem (Bechky & Okhuysen, 2011). In sum, reflexive learning may cause a team to revisit and update earlier assessments and actions as new knowledge emerges both in terms of the effectiveness of their actions and from the environment.

Phase 6: Knowledge and Skill Acquisition

The outcome of each adaptive performance episode can result in learning that has broader consequences for the team's ability to respond to adaptive stimuli through a "relatively permanent change in the team's collective level of knowledge and skill produced by the shared experience of the team members" (Ellis et al., 2003, p. 822). Learning as an outcome of the adaptive process through the acquisition of new knowledge and skills enables further growth and improvement (Maynard & Kennedy, 2016; Zellmer-Bruhn, 2003); thus, we place the knowledge and skill acquisition phase following the team's adaptive performance cycle. In our model, we propose that through reflection on adaptive experience and sensemaking, the team may be able to refine its general adaptive processes (Weick, 1995), as well as improve its overall level of "team adaptability" (e.g., Maynard et al., 2015) in terms of better strategies, higher collective efficacy, and higher quality mental models (Wilson et al., 2007). This link, therefore, suggests synergies between work on the learning phase of the adaptive processes and how it may relate to research on team attributes in isolation. Specifically, as shown in Figure 1, we draw a link from *Knowledge and Skill Acquisition* back to the beginning of our model, as the shared knowledge and experiences of the team should enhance each stage of the adaptation process for future stimuli, beginning with enhanced Detection.

Learning can also lead to positive outcomes from adversity, as teams may be able to apply their experience of a setback or failure to improve (e.g., Madsen & Desai, 2023). Failure is a "natural, instructive part of the learning process" (Unger-Aviram & Erez, 2016, pp. 241–242), and some researchers have argued that setbacks may also present opportunities for teams to learn, grow, and adapt, improving team performance (e.g., Cannon & Edmondson, 2005). The process of collectively making sense of a negative event through reflection may also allow teams to appraise the situation more objectively and avoid falling into cycles of conflict and blame (Rauter et al., 2018; Thomas et al., 1993). Further, if the team views the experience as an opportunity to learn, the shock to their routines may help them respond to future events with cognitive adaption and enhanced performance (Gersick & Hackman, 1990).

These views emphasize the critical role of experience in the learning process, where the team reflects on its adaptive efforts and learns from both successes and mistakes (e.g., Gibson & Vermeulen, 2003). For example, collective efficacy is built through shared experiences (Carmeli et al., 2013) and shared cognitions such as transactive memory and team mental models emerge through an interactive process of interaction and learning (Lewis et al., 2007; Pearsall et al., 2010). Some researchers have suggested that learning from failure, in particular, is essential for developing a mindset of preparedness (Carmeli & Schaubroeck, 2008), though others have found that improvement may also come vicariously by learning from the failures of others (e.g., Bresman, 2010; Kim & Miner, 2007); both may improve a team's adaptability. Learning from experience should also directly improve team adaptability by developing teams' ability to better detect, identify, and assess similar emergent stimuli, as their knowledge base and experience provide a broader understanding of their operational environment (Burke et al., 2006; A. Edmondson, 1999), to better cope with uncertainty, and lead to a better ability to balance proximity and magnitude in prioritizing among demands.

Knowledge and Skill Acquisition: Future Directions

An important area for further research is determining what factors within a team, or experiences, may help a team learn the "best" lesson from an adaptive episode. Prior research has suggested that teams tend to misclassify the reason for their performance and apply the wrong lessons to future adaptations (M. D. Johnson et al., 2013). Further, outcomes with negative valence tend to cause a loss of efficacy and withdrawal (e.g., Lazarus, 1991; Shepherd et al., 2013), making it difficult for teams to engage in beneficial sensemaking following a loss or failure. Some research has suggested that reflexive narratives may help teams sidestep the outcome itself and learn from the experience (Schippers et al., 2014) or that concept maps can help teams relearn after a change (Santos et al., 2021). Identifying specific factors that enable accurate learning from adaptive experiences is an important future direction.

Although we differentiate stimulus-based team adaptation from continuous improvement processes within the organization or team, the two learning processes may run parallel, or even intersect, in this phase. Teams may simultaneously need to continuously improve their outcome efficiencies through shared experiences and knowledge sharing across their primary responsibilities while separately developing specific "task mastery" within an adaptive domain to apply what they learn from each specific adaptive episode (A. C. Edmondson et al., 2007). These types of learning have been viewed as separate research streams, but have the potential to either enhance or interfere with one another. On the one hand, changes to structure or processes may necessarily come at the cost of disrupting efficient routines built over time (Gersick & Hackman, 1990). On the other hand, a deeper understanding through codified knowledge and smooth coordination through continuous improvement may also enhance teams' ability to respond to future unexpected demands. Future research connecting these streams, and examining factors that lead to synergy or conflict between them, is a valuable direction for researchers to explore.

We also suggest that another important avenue for future research revolves around clarifying the nature of the relationship between adaptation-based learning and the development of resilience in teams—a complex relationship that is currently unclear. In addition to member trait-level resilience, teams may also develop resilience as a capacity by working and learning together over time (becoming a *team attribute*) and recognizing they can successfully manage challenges. We argue that this connection to resilience can be integrated into models of team adaptation in the learning phase, as an outcome of "successful" adaptive responses, or even unsuccessful responses, if the teams feel as though they put forth their best effort.

Contextual Factors: Team Attributes and Structure

The ability of the team to effectively respond to an adaptive stimulus is also a function of the collective knowledge, skills, and characteristics of the team, including the way it is designed and structured. Some of these characteristics are the combined personality and cognitive ability of the members of the team, and others are the collective attitudes and cognitive understanding within the team, often derived from shared experiences and learning. The team's stage of maturity and development therefore acts as an important attribute affecting adaptive processes. The team's

structure includes the size, workflow, reward systems, and other task interdependencies within the team. Collectively, these characteristics of the team influence its level of team adaptability, defined by Maynard et al. as “the capacity of a team to make needed changes in response to a disruption or trigger” (2015, p. 655). Some studies of attributes and structure focus on team characteristics as antecedents of successful adaptation, while others explicitly examine them as moderators of other attributes or interventions.

Team Attributes Across Phases: Regulatory Focus and Goal Orientation as Exemplars

While many attributes may be differentially influential across the phases of our model, we wish to highlight two exemplars that we view as particularly influential in different phases to demonstrate the unique influence of attributes throughout the model: *regulatory focus* and *goal orientation*. Regulatory focus reflects the team’s orientation toward either the *prevention* of mistakes and errors or the *promotion* of the pursuit of new opportunities for gain, and its effects should be felt most keenly in the *early* phases of our model. A shared regulatory focus can prompt attention to different environmental demands, with promotion-focused teams on the lookout for opportunities, while prevention-focused teams diligently scan the environment for threats (P. D. Johnson et al., 2015). Those effects also pervade the *Urgency Assessment* phase, as the magnitude of losses may loom greater for prevention-focused teams than for those focused on gains. When aligned with the priorities of the organization, such a focus can be beneficial, but a misalignment may cause the team to mis-prioritize its threats and opportunities (Knight et al., 2001). The team’s reflexive learning processes may help remedy the urgency assessment, as they can challenge their own assumptions and biases as they update their assessment of the situation (Schipper et al., 2014).

In the later phases of the adaptation process, team goal orientation may play a larger role. A *learning goal orientation* can enhance the effectiveness of both *reflexive learning* and *knowledge and skill acquisition*. Teams with a shared learning orientation tend to view experiences as opportunities to improve (Bunderson & Sutcliffe, 2003), which is especially important when teams face setbacks and failures in their adaptation efforts. Teams with a learning orientation share a “climate of discovery and competence development” (Pearsall & Venkataramani, 2015, p. 737), engaging in greater information processing to understand the root causes of problems and how best to solve them. Learning orientation also helps teams respond to stimuli in the moment when they are assigned difficult goals (e.g., Han & Williams, 2008; LePine, 2005), whereas more *performance-oriented teams* with difficult goals were especially unlikely to adapt (LePine, 2005). Porter et al. (2010) reported a more complex picture of goal orientation. In teams with slack resources, learning orientation improved adaptive performance over time, whereas performance orientation was negatively related. Additionally, teams given an *outcome focus* were better able to address changes and outperformed teams that were given a *process focus* following a loss of materials (Woolley, 2009), possibly due to detecting threats sooner. Next, we turn to the more general benefits of team attributes in the adaptive process.

Other Team Attributes

Other attributes have been shown to have a more general benefit for team adaptation, and the roles of team member *cognitive ability* and *personality* have been most studied. Cognitive ability has been strongly related to adaptation behaviors, learning, mental model similarity and accuracy, information sharing, and performance (Ellis et al., 2003; LePine, 2003; Moon et al., 2004; Randall et al., 2011; Stokes et al., 2010). In terms of personality, *agreeableness* has been shown to be negatively related to learning in an adaptive context (Ellis et al., 2003). For *conscientiousness*, two facets, achievement and dependability, are conversely related to adaptive performance, such that role structure adaptation was higher for teams high in achievement and lower for teams high in dependability (LePine, 2003). *Emotional stability* has a direct impact on adaptation as teams with emotionally stable members were more likely to help other members (Porter et al., 2003), and it also enhanced the performance of teams when their leader was removed (DeRue et al., 2008). Abrantes et al. (2020) found that team member *temporal personality* may play a key role in dealing with unpredictable disruptions to the team’s external environment, as a *future orientation* among members enhanced improvised adaptation.

Other researchers have focused on collective attitudes and emergent states. Randall et al. (2011) found that *psychological collectivism* predicted information sharing, which influenced adaptation. *Psychological safety* has also been shown to play an important role in helping teams to effectively adapt when faced with change (A. C. Edmondson et al., 2001), and *self-efficacy* and *collective efficacy* positively impact adaptive performance (Chen et al., 2005; Randall, 2008) and performance when a newcomer joins a team (Chen, 2005). *Prevention climate* and *strategy* also play a moderating role, as groups adapting to member turnover performed worse when they were high in prevention climate or following a prevention strategy (Hekman et al., 2022).

Finally, the team’s maturity and development can influence how well it performs in each aspect of the adaptive process. Intact teams that have worked together for a long time may be much more effective at coordinating and understanding role linkages within the team (e.g., Kozlowski et al., 1999). They may identify and properly assess critical stimuli more quickly, based on extensive shared experiences and better understanding of one another’s capabilities. Similarly, they may be able to incorporate feedback from adaptive experiences into long-term learning better, as each new piece of information fits into their shared understanding built over time (Harvey et al., 2022). However, it is also possible that long-tenured teams may struggle to make needed changes, as members are used to acting in habitual ways within the team context (Gersick & Hackman, 1990).

Structure

There has also been support for the moderating role of structural factors in adaptation. An evenly distributed workload and a paired structure (when a member had a role partner who shared expertise and information processing responsibilities) have been positively related to team learning (Ellis et al., 2003). Additionally, consistent with structural adaptation theory, structural changes that move teams away from “team-ness” and toward individuality are more

effective than in the reverse manner (e.g., cooperative reward to competitive reward, functional to divisional structural shift, tightly coupled structure to loosely coupled structure, mechanistic to organic structures; Beersma et al., 2009; M. D. Johnson et al., 2006; Moon et al., 2004). For example, Hollenbeck et al. (2011) found it was more difficult for teams to shift from a loosely coupled structure (divisional structure, individual rewards, decentralized) to a tightly coupled structure (functional structure, group-based rewards, centralized).

Team Attributes and Structure: Future Directions

Most relevant to our new stimulus-based model, we see a need for future research that examples the impact of team attributes and structure across different phases of the team adaptation process. We have highlighted regulatory focus and goal orientation, but future work may find that certain team structures are best suited for urgency or duration assessment, or that while conscientiousness may positively influence stimulus detection, it negatively relates to flexibility and adaptive responses in later phases of our model. Another future question revolves around team resilience. *Team resilience* is a frequently referenced but often vaguely defined concept that overlaps with team adaptability. A common theme among conceptualizations of team resilience is the “capacity to bounce back from an adversity-induced process loss” (Stoverink et al., 2020, p. 397) or to withstand setbacks (Maynard et al., 2015). We see resilience potentially playing a moderating role in how well teams can mount an effective response to a change, similar to other forms of adaptability. Resilience may also play an important role in the later phases of our model, when teams must deal with setbacks and behaviors potentially not having the desired effect, and in terms of learning from their behaviors.

Another important area for future work involves a greater understanding of *external team contexts*, as discussed in a recent review (Ployhart et al., 2022). The authors discuss the concept of *munificence*, defined as “the portfolio of resources that exist in the external team context” (p. 1055), which can clearly impact how well a given team can adapt. We see a need for future work that integrates across this perspective—the resources available in the external team context—with work that examines internal team resources such as those discussed above, or slack resources in the team (e.g., Porter et al., 2010).

Contextual Factors: Leadership and Training

Leadership and training are means for the organization to increase the team’s adaptability prior to stimulus detection, or to directly affect the team’s adaptive response to a specific stimulus. These factors can therefore influence the adaptive process at multiple points. As with team attributes, we argue that while leadership will affect the entire team adaptation process, certain leadership elements should be most important in specific phases.

Leadership and Training Across Phases: Leadership as an Exemplar

Leadership attitudes and behaviors may shape the adaptive process differentially depending on the focal adaptive phase. During the *Stimulus Detection* phase, the leader may be perfectly positioned

to help scan the environment for emerging threats, opportunities, and demands while their team focuses on day-to-day performance. However, the leader’s attitudes and biases may create additional limits on the team’s ability to detect certain types of stimuli. As noted earlier, the leader’s goals and performance priorities may play a key role in prioritizing stimuli for their team through setting goals and monitoring their actions. For example, teams with *prevention-focused* leaders have been shown to overemphasize threat detection over identifying potential opportunities for gain (e.g., Pearsall et al., 2023). Similarly, the leader may play a key role in determining which stimuli are most urgent, based on their assessment of the magnitude of the demand. Once the adaptive response phase has begun, leader prevention focus may be less important, as the team has already prioritized and begun to attend to the demand.

In assessing urgency, *directive leadership* can be particularly helpful in reacting to some stimuli and more effective for adaptive team performance than empowering leadership (Sanchez-Manzanares et al., 2020) in high-magnitude crisis situations as it allows for a faster response (Yun et al., 2005). However, *empowering leadership* may be more effective prior to the detection of a stimulus as it enables team attributes such as shared cognitions and collective efficacy (Cheong et al., 2019) which may help members feel comfortable voicing stimulus detection behaviors. In later, *learning* phases, the explicit coordination provided by directive leaders may detract from team reflection, while the implicit coordination associated with empowering leaders should enhance team sensemaking about their adaptive efforts, helping them to update their earlier evaluations and revise their actions (Konradt et al., 2015; Lorinkova et al., 2013).

In total, the literature suggests that leaders can take an active role in helping the team detect and identify potential threats and opportunities in the environment, providing a broader external focus and helping make sense of the nature of new demands. When the stimulus is extremely urgent, in terms of proximity and magnitude, such as a disaster or crisis, the team may be best served by the clarity and speed of directive leadership. However, for less urgent situations, with long durations, such as the permanent loss of a member, and for learning processes, the team may be best off having control of how they restructure their work. We next turn to the general implications of leadership and training for the entire adaptive process.

Leadership and Training

Most of the limited research on leadership in team adaptation has focused on how leaders can better help teams effectively respond to demands. *Leader sense-giving* and *briefings* about strategies have been shown to be particularly important in improving team mental models to improve adaptive performance (Marks et al., 2000; Randall et al., 2011). Leaders can also directly help teams adapt by providing active *coaching*, communicating, and reinforcing the adaptive plan, and helping the team to see the broader implications of their work (A. C. Edmondson et al., 2001; Morgeson, 2005). *Supportive and active coaching*, as well as *sensemaking*, can also help build team trust in the leader and have been shown to enhance perceptions of leader effectiveness in the face of a novel event (Morgeson, 2005).

Informal, or shared, leadership, in which the team’s leadership responsibilities are shared among team members (Hiller et al., 2006), has been shown to be positively related to routine team

performance and extreme action teams (Carson et al., 2007; K. J. Klein et al., 2006) and has been suggested to be beneficial for adaptation (Burke et al., 2006). However, the little work that has examined its direct role in team adaptation has been inconclusive, as Koeslag-Kreunen et al.'s (2018) meta-analysis indicated that *vertical team leadership* was more suited than shared leadership in enabling team learning behaviors in tasks that required adaptation.

Organizations can also prepare their teams for adaptation more indirectly through training interventions. Researchers have shown that teams receiving (a) *adaptive training* or (b) *adaptive training combined with mission briefings* performed better than control teams under stressful conditions and exhibited higher teamwork processes (Entin & Serfaty, 1999). *Team interaction training, concept mapping cross-training, self-training and development, and perturbation training* (where standard procedures are disrupted during task acquisition, forcing teams to coordinate in novel ways) positively impact mental models, communication, adaptive behaviors, and team adaptive performance (Chandler & Lyon, 2009; Gorman et al., 2010; M. D. Johnson et al., 2013; Marks et al., 2000; Santos et al., 2021). Researchers have also found support for *formal interventions* related to member familiarity in promoting adaptation (Okhuysen, 2001). For example, teams that completed a *team charter* performed better in an adaptive task, mediated by their ability to handle disruptions (Sverdrup et al., 2017).

Leadership and Training: Future Directions

Given the increased interdependence of work teams (cf. Cohen & Bailey, 1997), it is important for future research to also examine how leaders can allow teams to successfully adapt to the complexities of high team and task interdependence, while also adapting to changing goals and priorities. This area may be informed by research on larger scale organizational interventions, such as team building, which finds that not only do these interventions improve some measures of performance (C. Klein et al., 2009), but they also impact team attitudes such as cohesion (Senécal et al., 2008).

Preemptive adaptability training may also prove valuable for teams. At the individual level, Pulakos et al. (2000) focused on the role of purposeful training and skill enhancement designed to improve future adaptive performance. Their work suggests that a key aspect of adaptability should be “anticipating changes in the work demands and searching for and participating in assignments or training that will prepare (the) self for these changes” (p. 617). Teams that spend time preparing for future demands may be able to detect them sooner, better assess their urgency, and adapt with a specific and practiced response.

Methodological Challenges in Studying Team Adaptation and Research Agenda From a Stimulus-Based Perspective

Based on our stimulus-based phase model of team adaptation, we next identify methodological trade-offs inherent in studies of team adaptation and provide recommendations for study design and a prioritized agenda for future research in this area. Adaptation researchers face numerous methodological challenges and must make important choices when conceptualizing and measuring team adaptation. Those decisions can lead to very different findings and can make drawing inferences from, and comparing between, various

studies challenging. For instance, some research views adaptation in terms of total team performance following a change (Marks et al., 2000), but others measure it through strategic choices (e.g., Randall et al., 2011), adaptive behaviors (e.g., LePine, 2003), or the timing of those behaviors (e.g., Waller, 1999), and it is unclear whether modifications in cognitive structures or behavioral routines are needed to demonstrate successful adaptation.

We propose a specific research agenda for future work that highlights these methodological challenges and prioritizes the most critical issues to address. Based on our process model, we suggest researchers focus their attention on five major issues: measuring successful adaptation, team temporal focus, stimulus detection, team development, and team adaptive/performance capacity.

The first, most critical issue is identifying when, or measuring how much, a team has successfully “adapted.” As we discuss in *Phase 4: Adaptive Response and Performance*, this issue often boils down to behaviors versus outcomes. It is still unclear whether the more important concern is the process by which a team adapts (i.e., adaptive behaviors) or the result of its efforts (i.e., adaptive performance). This issue becomes even more complex when attempting to differentiate general effective team performance from adaptive performance specifically. Our stimulus-based perspective can provide some insight into this issue as it places a focus on what exactly the team must adapt to, and what is needed to face, rather than beginning with general team inputs that can be helpful for team performance in general.

We, therefore, recommend that future researchers interested in disentangling adaptation from general performance focus even more on the idea of *stimulus-specific actions*, defined as “actions or behaviors enacted by the team that directly address, and are limited to, a particular and discrete demand” (Christian et al., 2017, p. 65). These behaviors—such as tailored communication workarounds in the context of a breakdown (LePine, 2003) or sharing relevant new information learned during a team action phase (Ellis et al., 2003)—are unique to a discrete demand and thus can help provide clarity around how well the team has adapted to a particular stimulus.

As field studies have begun to complement laboratory research on team adaptation, researchers have begun addressing the issue of assessing effective adaptation using novel situations and new measures. For example, Stephens (2021) focused on assessing adaptation through the overall “aesthetic experience” of choir members, and other recent field work has focused on emergent response teams. To capture emergency situations in real time, some researchers have employed realistic scenario tests, such as Schmutz et al.'s (2018) study of medical teams in which they recorded and coded team responses to four different emergency situations. In such cases, successful adaptation may be dictated so heavily by internal measures of success that external researchers lack the experiential understanding of the challenges teams face and have therefore had to rely more heavily on qualitative or ethnographic work (e.g., Geiger et al.'s, 2021, study of firefighters; Bechky & Okhuysen's, 2011, study of special weapons and tactics teams and film crews).

Similarly, responses to a completely unexpected stimulus, such as an environmental disaster, can only be examined after the fact, and researchers have had to employ novel methods to capture team adaptation in such settings. For example, in examining a team's “radical and swift” reorganization in response to Hurricane Dorian, researchers employed a combination of interviews with team

members, photographs, and videos, including many scraped from social media following the relief effort (Kalkman, 2023). Web-scraping and social media posts may be a new frontier in field-based team adaptation examinations. As researchers attempt to examine real-world teams in complex, uncertain situations, we recommend that they ground their work in the specific problems and measures of their setting through such interviews and post hoc sensemaking which may be instrumental for identifying stimulus-specific actions unique to the particular demand, and within each specific phase of adaptation.

Second, research should focus more on the issue of time in team adaptation, particularly in the temporal focus on the team. Our phase model also suggests methodological insights from teams' expectations of the duration of the adaptive stimulus, for which the team designs its broader approach. For a research question involving short-term responses to temporary changes, behavior may be a much better indicator of effective adaptation than collective performance outcomes, which may take time to compile. For teams facing more sustained changes, the distal performance implications of their actions may provide much greater insight into the effectiveness of their adaptive response as future performance may matter more than short-term reactions, and the team also has more time to engage in reflexive learning and updating where they alter ineffective courses of action.

The issue of temporal focus is particularly salient because the relationship between individual member behavior and collective performance may not be obvious and may not manifest immediately, and many effective adaptive responses in the long term may have short-term performance costs. For example, a team that responds to changes in the environment or in competition may need to invest considerable time and effort in external scouting (D. G. Ancona & Caldwell, 2007), learning new skills and abilities, or recruiting and training new members. Ultimately, these adaptive responses may pay off, but in the short term, they may be viewed as counterproductive. Similarly, a team that responds to competitor innovation by increasing internal efficiencies may see a short-term boost in performance, but ultimately fail because their adaptive response was incorrect over a longer horizon.

Issues of time have received recent attention in the literature (e.g., Geiger et al., 2021; Kooij et al., 2018), highlighting that some situations present temporal uncertainty—a “fundamental uncertainty about when crucial events or actions are going to occur” (Geiger et al., 2021) and that temporal misfits can lead to organizational failures. Even when a stimulus may appear straightforward, team members may not agree about the current state of affairs and what is likely to happen in the future (Kaplan & Orlikowski, 2013), have different levels of construal related to how much they see the stimulus as affecting the “here and now” (construal level theory; Trope & Liberman, 2003), or may have trouble accurately diagnosing the duration of the stimulus in Phase 3 of our model. Given issues in terms of accuracy in time perception and difficulties in forecasting future events (e.g., Weick & Sutcliffe, 2015), we recommend that researchers investigate the question inherent in *Duration Assessment*: How accurately can teams assess the duration of a stimulus at its outset and come to a shared assessment, and following this, how do they incorporate novel information to update their duration assessment? As we noted in discussing *reflexive learning*, the updating process is critical for teams to continue to learn and improve within an adaptive cycle, and research is needed that

examines the assessment and updating process over time. Researchers employing longitudinal designs have implicitly understood this issue. For example, Konradt et al. (2015) explored the role of reflection in adaptation over time, showing that, as teams received feedback during adaptation and updated their team and task mental models, adaptive performance improved.

Third, the extant literature has generally neglected the Detection phase of the adaptation process, and our stimulus-based perspective suggests a need for more attention and measurement in the early phases of our model, particularly in the stimulus detection phase. As noted in our discussion of Phase 1 of our model, *Stimulus Detection* may be fraught with “informational ambiguity” (Beersma et al., 2016) and researchers have not developed a clear definition or measure for when detection occurs. It is not enough for a member to sense a problem; the rest of the team must be informed and then collectively identify it as a stimulus. Most laboratory studies have employed stimuli with unmistakable demands or the obvious loss of member (e.g., DeRue et al., 2008; Kaplan et al., 2013), ensuring that teams in the study all recognize the stimulus at the same time so that researchers can isolate the effectiveness of their response. Few researchers have explicitly examined how quickly detection occurs and those have tended to employ simulations. For example, Schmutz et al. (2018) had medical teams assess patients with unclear conditions to see how quickly they could detect and identify the specific problem and respond to it. Waller (1999) employed a training simulator with an unexpected emergency for flight crews and determined that detection had officially occurred only when it was verbalized by a member to the rest of the team.

In other fields, emergency response organizations have approached this problem by developing detection technologies. For example, researchers have examined the effectiveness of ShotSpotter technology, which employs acoustic sensors positioned to detect gunshots, allowing police and emergency response teams to deploy to the location of a shooting much faster. This faster detection has resulted in shortened transport times for victims to trauma centers and provided medical teams more time to respond to the injury (Goldenberg et al., 2019).

We recommend that researchers designing measures of stimulus detection should consider focusing on behavioral measures (e.g., action was taken toward the stimulus specifically) or team member conversations and discussions (e.g., email identifying a potential opportunity sent to other team members). A focus on this issue can also lead to investigations of novel questions that follow from the overall detection, such as how quickly certain stimuli tend to be detected. Decades of research on groups and teams show that team members do not always share all information and may deem certain pieces of information as unimportant (e.g., Stasser & Titus, 1985); therefore, is there a “critical mass” of members that must detect a stimulus before the team moves on to Phase 2 in our model?

Then, once a stimulus is detected, how can team members reconcile potential differences in their personal assessments of urgency based on proximity and magnitude? As noted by Kaplan and Orlikowski (2013), team members often have “different projections of what could take place” (p. 965) and even differ in terms of their view of current issues and changes. Theories of team decision making (e.g., Hollenbeck et al., 1995) point to the importance of both properly weighting individual member's perspectives based on their area of expertise, and then having that information distributed throughout the team. In particular, the ability of the team to

incorporate minority viewpoints that run counter to team expectations can be critical for identifying uncommon stimuli or avoiding stimulus identification mistakes (Park & Deshon, 2010). We recommend that researchers seek to directly measure whether teams in the field detected a stimulus in the first place, and how they decided to prioritize various stimuli in terms of proximity and magnitude. Such designs could include both longitudinal within-team studies over time, or shorter term teams in either laboratory or field settings.

Fourth, we also recommend that researchers focus attention on the stage of maturity and development of the team as an important contextual factor when they face an adaptive stimulus. As noted earlier, team tenure and maturity can be an important attribute that shapes their reactions, positively or negatively, to each stage of the adaptive process. Studies in the field or in longer term classroom settings could provide important insights into how experienced teams employ knowledge and skills acquired in previous adaptive cycles (Phase 6) to new demands. Further, as adapting to a stimulus is a process that may take some time, studies that address additional stimuli that occur *while* the team is in the process of adapting would be valuable. Certain stimuli, for example, an internal disruption such as a change in CEO, may coincide with external demands such as an attack by a competitor (Lee & Tsai, 2023). Qualitative recollections of such experiences from existing teams may provide insight into key team traits and attitudes, and laboratory studies may be able to capture behaviors in a controlled setting to examine if and when teams are able to detect a new stimulus when engrossed in a current challenge, and identify which factors help teams to leave their attention open wide enough to address both changes as they unfold.

Finally, many researchers use the team's initial, routine, performance as a baseline from which they assess the team's ability to recover. Teams that initially perform well have more capacity to decline, simply based on their high initial performance. Poor performing teams, therefore, may find it much easier to successfully "adapt" back to their initial starting point since it was lower in the first place, making the change in performance during adaptation a potentially misleading value. In such cases, it may make the most sense for researchers to focus on the adaptive behaviors that result from the recognition of a stimulus by the team rather than its actual recovery in performance (e.g., LePine, 2003, 2005). Alternatively, researchers may choose to include prior performance in all analyses (see DeRue et al., 2008; Randall et al., 2011). This will further help isolate the effects of team behaviors and states specific to adaptation rather than general attributes of the team that are beneficial across contexts.

In sum, our model suggests a focus on more targeted adaptive responses—those stemming from proper assessment in earlier phases—which can help provide even more nuance around effective adaptive behaviors and provide a more fine-grained analysis of which behaviors are key in responding to different stimuli.

Conclusion

Factors enhancing the ability of teams to successfully adapt have been increasingly studied over the last 20 years, but thus far, the disjointed nature of the research has made understanding and applying the resultant knowledge difficult. Our stimulus-based integrative conceptual review of the team adaptation literature employs a multiphase process model to help understand the factors that influence teams' ability to proceed through each phase and the

downstream consequences of that process. Further, we identify and review several critical factors related to the stimulus itself, the attributes and structure of the team, and leadership and training.

In doing so, we make several important contributions to the team adaptation literature: We provide a theoretical framework for understanding the team adaptation process that begins with the adaptive stimulus, we challenge the extant literature by adding continuity to the attributes of the stimulus that have been often treated as categorical, we identify critical contextual factors and boundary conditions of the relationship between assessment and the team's behavioral response, we describe how team attributes and leadership may differentially impact phases of the adaptive process, and we extend our model to highlight new and emerging directions for further study. Finally, we also detail methodological challenges unique to studying team adaptation and provide a prioritized research agenda from a stimulus-based perspective.

We hope that our model has helped to provide insight to researchers and encouraged them to think more deeply about the type of adaptation they wish to study and how to position their research within our phase model. We also hope that our model helps researchers to recognize critical similarities and differences in the types of adaptation studied in the literature, and important areas in need of future research.

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